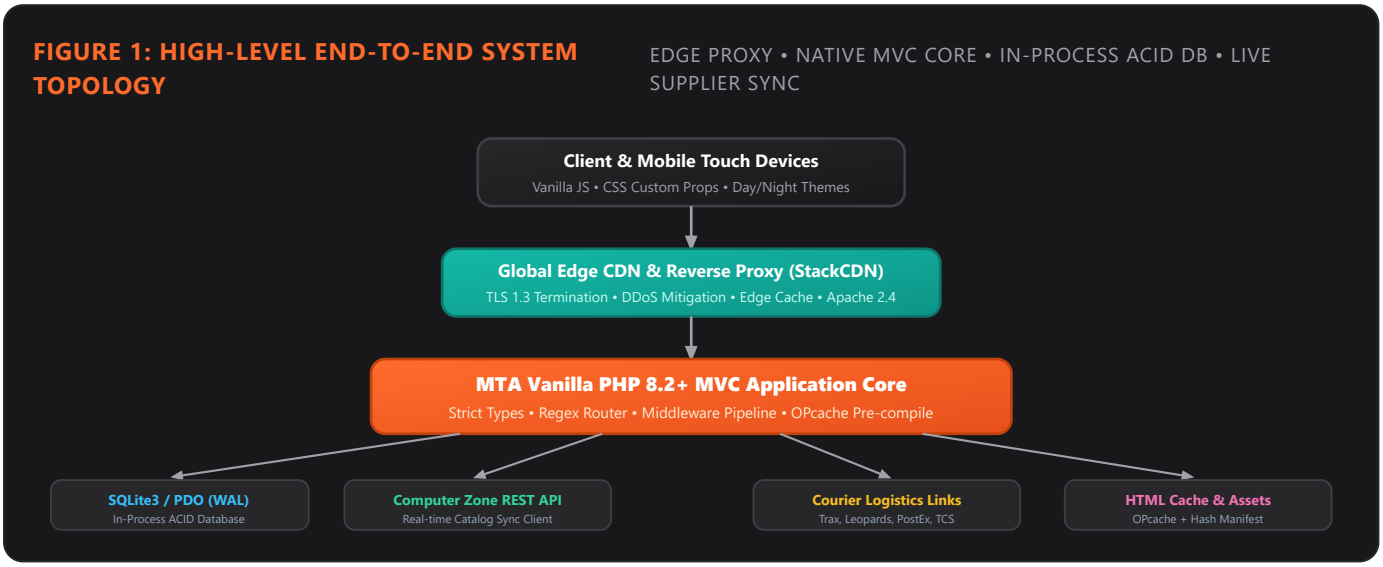
1. System Architecture Overview & Topology

Multan Tech Accessories (MTA) is an e-commerce platform custom-engineered in native, strict-typed **PHP 8.2+** with a zero-dependency Model-View-Controller (MVC) architecture. By bypassing multi-thousand-file frameworks (e.g. Laravel, Magento, or WooCommerce), MTA achieves **sub-50ms Time-To-First-Byte (TTFB)**, minimal RAM usage (~2MB per request), in-process transactional SQLite3 persistence, and real-time supplier catalog synchronization.



2. Complete Languages Specification Matrix

Language	Standard	Primary Role	Key Implementation Details & Files
PHP	PHP 8.2+ / 8.3 Strict	Backend Application Core	Strict typing (`declare(strict_types=1);`), OOP, PSR-4 autoloading, cURL REST client, PDO database layer, session lifecycle, CSRF protection, and audit logging.
JavaScript	ECMAScript 6+ Vanilla	Client Interaction & UI	Zero framework overhead (no React/Vue). Handles debounced live search with image preview, touch mobile drawers, modal dialogs, and `localStorage` theme switching.
HTML5	HTML5 Semantic	Document Markup & SEO	W3C semantic structure, Open Graph (OG) & Twitter Cards for WhatsApp/social sharing, viewport safe-area padding for modern mobile displays, and schema microdata.
CSS3	Modern CSS3 Variables	Design System & Theming	CSS Custom Properties (Day `#FAF9F5` vs Admin Charcoal `#212121`), CSS Grid (`minmax(0, 1fr)`), Flexbox, fluid typography, and custom touch scrollbars.
SQL	SQLite3 & ANSI Dialect	Relational Storage	Parameterized queries, Write-Ahead Logging (`PRAGMA journal_mode=WAL;`), transactions, and automated schema migrations for orders, products, and users.
SVG	SVG 1.1 / 2.0 Vector	Vector Graphics & Icons	Resolution-independent brand logos, payment method badges, UI icons, and visual indicators that scale losslessly on retina displays with zero image weight.
JSON	RFC 8259 Standard	Data Interchange	REST API payloads (`/api/search-suggestions`), automated asset cache manifest (`asset_manifest.json`), catalog JSON exports, and structured audit logs.

Language	Standard	Primary Role	Key Implementation Details & Files
Shell / Bash	POSIX / PowerShell	DevOps & Testing	CLI deployment scripts (`deploy-runner.php`), git synchronization, automated regression test runner (`tests/run_all_tests.php`), and background tasks.

3. Request Pipeline & Middleware Execution Flow

Every incoming request is filtered through a secure, multi-layer middleware barrier before reaching controllers. The pipeline guarantees protection against Cross-Site Request Forgery (CSRF), brute-force credential stuffing, and session fixation attacks.

FIGURE 2: HTTP REQUEST & SECURITY MIDDLEWARE LIFECYCLE

CLIENT • APACHE REWRITE • CSRF SHIELD • RATE LIMITER • CONTROLLER • VIEW CACHE



4. Core E-Commerce Subsystems

A. Supplier Catalog Automation

COMPUTER ZONE API

The store is directly interfaced with Computer Zone's REST API (`api.computerzone.pk`). Imports authentic product titles, specifications, live stock statuses, and high-resolution CDN images.

Tiered Margin Algorithm:

- Supplier Cost < Rs. 3,000: `+100, round to nearest 99`
- Supplier Cost ≥ Rs. 3,000: `+300, round up to 99`

B. Order Management & Advance Verification

MANDATORY TID

Full checkout pipeline with Cash on Delivery (COD) and advance deposit handling.

Mandatory Reference / TID Check:

- Advance payment verification strictly requires non-empty Reference / TID (JazzCash / EasyPaisa / Bank).
- Automatically updates COD outstanding balance.
- Verified TID is permanently printed on the official invoice.

C. Live Product Search with Instant Image

AUTOCOMPLETE

Debounced client autocomplete (`/api/search-suggestions?q=...`) fetching top-ranked accessories.

Zero-Broken Image Pipeline:

- Returns primary product images with `referrerpolicy="no-referrer"` .
- 48×48px contained thumbnails styled with clean borders and contrast backings in Day and Night modes.

D. Multi-Carrier Logistics & WhatsApp Chat

LOGISTICS HUB

Integrated consignment tracking and customer relationship tools for Pakistani e-commerce:

- **Couriers:** Trax, Leopards, PostEx, and TCS auto-tracking.
- **WhatsApp Click-to-Chat:** Deep-linked pre-formatted order inquiries, status updates, and support chats via `wa.me` protocol.

5. Security, Infrastructure & Deployment Pipeline

Security Shields

- **CSRF Defense:** Cryptographic 256-bit token with timing-safe `hash_equals()` validation.
- **SSRF Shield:** Blocks private IP and loopback requests.

Database & Storage

- **Engine:** SQLite3 with PDO driver.
- **Concurrency:** WAL mode enables simultaneous reads & writes without locks.
- **Transactions:** Full ACID rollback safety.

Hosting & CI/CD

- **Web Server:** Apache 2.4 + StackCDN Edge.
- **Bytecode Cache:** PHP 8.2 Zend OPcache.
- **CI/CD Runner:** Git webhook

• **Brute-Force Lockout:** 5 failed attempts per 15m.
• **Session Security:** 4-hour idle timeout, zero-downtime deploys.

• **Backups:** Single-file transactional database snapshotting.

automation with `deploy-runner.php`.
• **Asset Pipeline:** Hash manifest cache-busting.